A RETRIEVAL COMPANION FOR SP 21

BIS *Compass.*

Find your Indian Standard in seconds, not weeks.

BIS STANDARDS IN SP 21	559
PAGES OF REGULATION	929
TYPICAL MSE SEARCH TIME	~3 WKS
AI TOOLING AVAILABLE TODAY	0

Indian Micro & Small Enterprises spend weeks searching the BIS catalogue to identify which standards their products must comply with. The 929-page SP 21 alone summarises 559 standards across building materials — discovery is manual, error-prone, and a bottleneck for getting products to market.

THE PRODUCT

Describe your product. *Get the right BIS standards.*

A retrieval-augmented engine purpose-built for SP 21 building materials. Plain-English in, ranked IS codes out — every recommendation grounded in the corpus, every code verified against a whitelist of real standards.

01 — Speed

Fast.

Sub-second retrieval

Hybrid sparse + dense recall, GPU-accelerated reranking. `~0.5s` average on consumer GPU — well under the 5s target.

02 — Quality

Accurate.

100% Hit Rate @3

On the public test set, every query returned the correct standard within the top three results. `MRR @5 = 0.93`.

03 — Trust

Grounded.

Zero hallucinations

Every recommended IS code is checked against a whitelist extracted from SP 21 itself. No invented standards. Ever.

SYSTEM OVERVIEW

End-to-end RAG pipeline

Built for sub-second latency on consumer hardware — from PDF ingest to grounded answer in six stages.

01 / Ingest

Ingest

PyMuPDF
559 records
parsed from SP 21

→

02 / Index

Index

FAISS
BM25
ColBERT
tri-modal recall

→

03 / Fuse

Fuse

Reciprocal Rank Fusion
1 / (60 + rank)
parameter-free

→

04 / Rerank

Rerank

bge-reranker-v2-m3
cross-encoder
top-25 → top-5

→

05 / Guard

Guard

IS-code whitelist
grounding check
zero hallucinations

→

06 / Serve

Serve

FastAPI
Next.js
Top-5 + rationale

EMBEDDER

BAAI/bge-m3

1024-d · fp16 · GPU

RERANKER

bge-reranker-v2-m3

cross-encoder

VECTOR STORE

FAISS · IndexFlatIP

cosine via inner-product

LLM (UI ONLY)

Gemini 2.0 Flash

rationale rewrite · optional

WHY HYBRID RETRIEVAL WORKS

Each component handles a failure mode the others miss.

CHUNKING STRATEGY

One chunk per standard

We segment the 929-page PDF on the recurring `'SUMMARY OF IS XXXX : YYYY <TITLE>'` header. Each chunk preserves the IS code, title, scope, revision, and first 1000 chars of body — capturing both the dense semantic core and exact technical vocabulary.

EMBEDDING TEXT · SAMPLE

```
IS 269: 1989 — ORDINARY PORTLAND CEMENT, 33 GRADE
Scope: Manufacture and chemical and physical
requirements of 33 grade ordinary Portland cement…
↳ embedded as a single 1024-d vector
```

1 BM25 sparse

rank_bm25 · title + scope + body

Catches rare technical vocabulary like `M30`, `OPC33`, `mortice`.

2 bge-m3 dense

1024-d embeddings · FAISS IP

Captures paraphrased and colloquial product descriptions — what an MSE owner actually types.

3 RRF fusion

1 / (60 + rank) · no learned weights

Parameter-free combination. Robust across query distributions without tuning.

4 Cross-encoder rerank

bge-reranker-v2-m3 · top-25

Explicit query↔passage attention fixes ranker mistakes the bi-encoder couldn't.

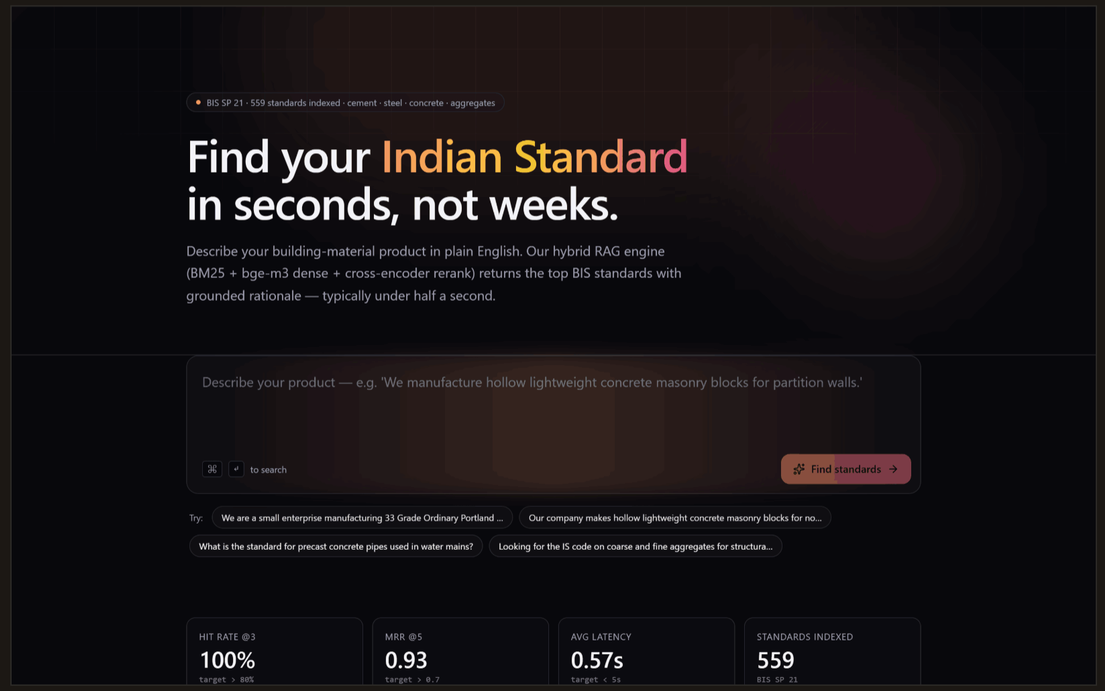
PRODUCT WALKTHROUGH

What an MSE owner sees.

Live screenshots from the production Next.js demo — query in, ranked top-5 BIS standards out.

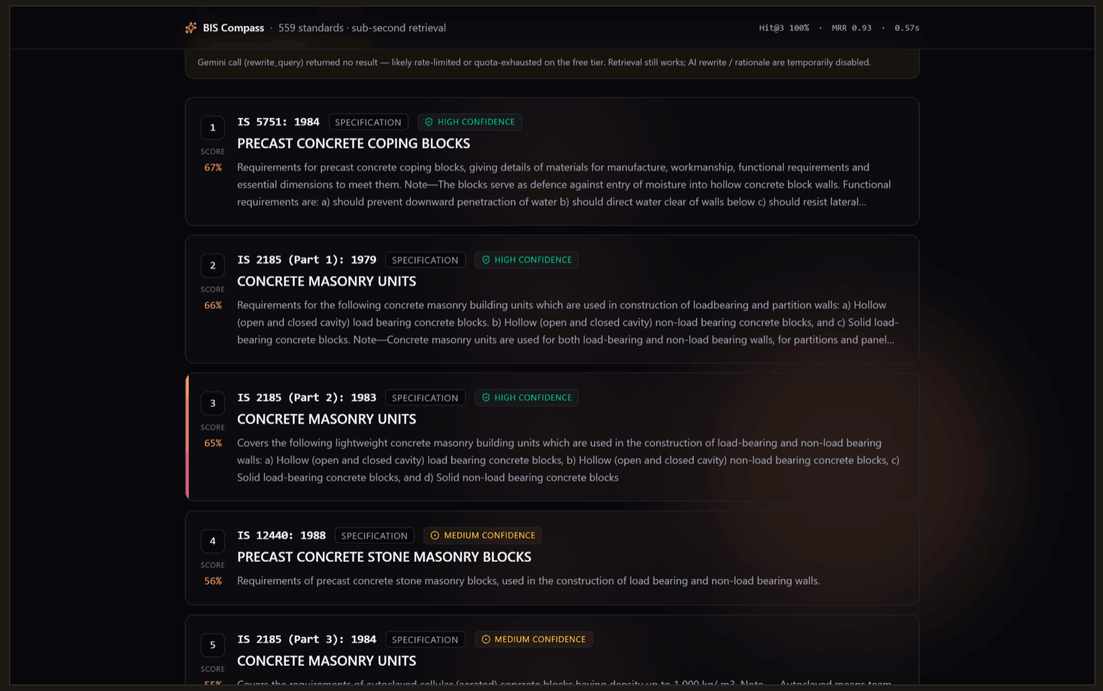
INPUT · PLAIN-ENGLISH QUERY

SCREEN 01



OUTPUT · RANKED TOP-5 BIS STANDARDS

SCREEN 02



END-TO-END LATENCY

<1 s

STANDARDS INDEXED

559

HALLUCINATED CODES

0 / 140

TARGETS VS. MEASUREMENTS

All three targets — *beaten by 2× to 11×.*

HIT RATE @3	MRR @5	AVG LATENCY
100%	0.93	<1s
target > 80%	target > 0.7	target < 5s
1.25× target	1.33× target	GPU · CPU OK

ABLATION · MRR @5 ON BOOTSTRAP SET

Hybrid + rerank is our production stack. Adding more priors regresses both sets.

EVAL SETS COMPARED

EVAL SET	HIT @3	MRR @5	LATENCY
public_test_set.json	100.00%	0.9333	0.45s
bootstrap_test_set.json synthesised, 18 queries	88.89%	0.9028	0.55s

FROM 3 WEEKS → 0.5 SECONDS

From weeks of regulatory search to *half a second.*

BEFORE · STATUS QUO

~~Manual catalogue search~~

- × Three weeks average to identify applicable standards
- × Browsing a 929-page PDF catalogue manually
- × Misses adjacent and cross-referenced standards
- × No way to triage across product variants

→

AFTER · BIS COMPASS

Sub-second discovery

- → Top-5 with confidence scores in <500 ms
- → Plain-English query → ranked IS codes
- → Hybrid retrieval surfaces related standards
- → Grounded rationale citing the actual scope text

Downstream effect

Faster compliance discovery → quicker product launches → more BIS-certified MSEs in the market. Multiplied across thousands of small manufacturers in cement, steel, concrete, and aggregates, this compresses an industry-wide bottleneck from quarters to days.

TEAM SIGMA SQUAD

Built solo, *in seven days.*

OPEN-SOURCE STACK

Standing on giants.

Retrieval

SEARCH CORE

MODELS
BAAI/bge-m3
bge-reranker-v2-m3

INDEX
FAISS
rank_bm25

COMPUTE
PyTorch 2.11
cu128 · fp16

TOOLING
transformers
sentence-transformers

Data & ship

PDF → API → UI

INGEST
PyMuPDF

API
FastAPI
Uvicorn

LLM (UI)
Gemini 2.0 Flash
google-genai

FRONTEND
Next.js 16 · Tailwind v4
Framer Motion · shadcn

Thank you. Bureau of Indian Standards · Sigma Squad organisers · Beijing Academy of AI (bge-m3, bge-reranker-v2-m3) · Google AI Studio · Hugging Face Hub · the open-source maintainers behind FAISS, PyTorch, FastAPI, and Next.js.